

寒假练习 05

一. 填空题.

1. $\frac{1}{x} + C$.

析: $\int \frac{f'(hx)}{x} dx = \int f'(hx) dhx = \int df(hx) = f(hx) + C = \underset{\substack{\uparrow \\ f(x)=e^x}}{e^{-hx}} + C = e^{\frac{1}{x}} + C = \frac{1}{x} + C$.

2. $-\frac{\sqrt{1-x^2}}{x} \arcsin x + \ln|x| + \frac{1}{2}(\arcsin x)^2 + C$.

析: 令 $\arcsin x = t$. ($-\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$). 则 $x = \sin t$, $dx = \cos t dt$

$$\begin{aligned} \int \frac{(1+x^2) \arcsin x}{x^2 \sqrt{1-x^2}} dx &= \int \frac{(1+\sin^2 t) \cdot t}{\sin^2 t \cdot \cos t} \cos t dt = \int t \csc^2 t dt + \int t dt \\ &= -\int t d\cot t + \frac{1}{2}t^2 = -t\cot t + \int \frac{\cot t}{\sin^2 t} dt = -t\cot t + \ln|\sin t| + C + \frac{1}{2}t^2 \\ \therefore x = \sin t, \therefore \cos t = \sqrt{1-x^2}, \cot t &= \frac{\sqrt{1-x^2}}{x}. \end{aligned}$$

$\therefore$ 原式 $= -\frac{\sqrt{1-x^2}}{x} \arcsin x + \ln|x| + C + \frac{1}{2}(\arcsin x)^2$

3. $-\cot x - \tan x + C$.

析: $\int \frac{\cos 2x}{\cos^3 x \sin^3 x} dx = \int \frac{\cos^2 x - \sin^2 x}{\cos^2 x \sin^2 x} dx = \int (\sec^2 x - \csc^2 x) dx = -\cot x - \tan x + C$.

4. $\frac{2}{9}(x^3+1)^{\frac{3}{2}} + C$.

析: $\int x^2 \sqrt{x^3+1} dx = \frac{1}{3} \int \sqrt{x^3+1} dx^3 = \frac{1}{3} \int (x^3+1)^{\frac{1}{2}} d(x^3+1) = \frac{1}{3} \times \frac{2}{3} (x^3+1)^{\frac{3}{2}} + C$.

5. $-\frac{1}{2\sqrt{3}} \arctan \frac{\sqrt{3} \cot x}{2} + C$ 或 $\frac{1}{2\sqrt{3}} \arctan(\frac{2}{\sqrt{3}} \tan x) + C$.

析: $\int \frac{dx}{3+\sin^2 x} = \int \frac{\frac{1}{\sin^2 x} dx}{\frac{3}{\sin^2 x} + 1} = \int \frac{\csc^2 x dx}{3\csc^2 x + 1} = \int \frac{d(-\cot x)}{3\cot^2 x + 4} = \frac{1}{4} \int \frac{d(-\cot x)}{(\frac{\sqrt{3}}{2} \cot x)^2 + 1}$
 $= -\frac{1}{4} \int \frac{\frac{2}{\sqrt{3}} d(\frac{\sqrt{3}}{2} \cot x)}{(\frac{\sqrt{3}}{2} \cot x)^2 + 1} = -\frac{1}{2\sqrt{3}} \arctan \frac{\sqrt{3} \cot x}{2} + C$.

或: $\int \frac{dx}{3+\sin^2 x} = \int \frac{\frac{1}{\cos^2 x} dx}{\frac{3}{\cos^2 x} + \tan^2 x} = \int \frac{\sec^2 x dx}{3\sec^2 x + \tan^2 x} = \int \frac{d \tan x}{3(\tan^2 x + 1) + \tan^2 x}$
 $= \int \frac{d \tan x}{4\tan^2 x + 3} = \frac{1}{3} \int \frac{d(\tan x)}{(\frac{2}{\sqrt{3}} \tan x)^2 + 1} = \frac{1}{3} \int \frac{\frac{\sqrt{3}}{2} d(\frac{2}{\sqrt{3}} \tan x)}{(\frac{2}{\sqrt{3}} \tan x)^2 + 1}$
 $= \frac{1}{2\sqrt{3}} \arctan(\frac{2}{\sqrt{3}} \tan x) + C$.

6. $\underline{-2 \arctan \sqrt{1-x} + C}$

析: 令 $\sqrt{1-x} = t$, ($t \geq 0$), 则 $x = 1-t^2$, $2-x = 1+t^2$, $dx = -2tdt$.

$$\int \frac{dx}{(2-x)\sqrt{1-x}} = \int \frac{-2tdt}{(1+t^2) \cdot t} = -2 \int \frac{dt}{1+t^2} = -2 \arctan t + C = -2 \arctan \sqrt{1-x} + C.$$

7. $\underline{x^3 + \arctan x + C}$

析: $\int \frac{3x^4 + 3x^2 + 1}{x^2 + 1} dx = \int \frac{3x^2(x^2+1) + 1}{x^2+1} dx = \int (3x^2 + \frac{1}{x^2+1}) dx = x^3 + \arctan x + C.$

8. $\underline{\frac{2}{\sqrt{\cos x}} + C}$

析: $\int \frac{\tan x}{\sqrt{\cos x}} dx = \int \frac{\sin x dx}{(\cos x)^{\frac{3}{2}}} = -\int (\cos x)^{-\frac{3}{2}} d\cos x = \frac{2}{\sqrt{\cos x}} + C.$

9. $\underline{-\frac{1}{4}x \cos 2x + \frac{1}{8} \sin 2x + C}$

析: $\int x \sin x \cos x dx = \frac{1}{2} \int x \sin 2x dx = -\frac{1}{4} \int x d\cos 2x = -\frac{1}{4} x \cos 2x + \frac{1}{4} \int \cos 2x dx$
 $= -\frac{1}{4} x \cos 2x + \frac{1}{8} \sin 2x + C.$

10. $\underline{\frac{1}{2}(\ln x)^2}$

析: 在 $f'(e^x) = x e^x$ 中, 令 $e^x = t$, 则 $e^x = \frac{1}{t}$, $x = \ln t$, $f'(t) = (\ln t) \cdot \frac{1}{t}$.

$\therefore f'(x) = \frac{\ln x}{x}$, $\therefore \int f'(x) dx = \int \frac{\ln x}{x} dx = \int \ln x d(\ln x) = \frac{(\ln x)^2}{2} + C$

又 $f(1) = 0$, $\therefore C = 0$.

$\therefore f(x) = \frac{(\ln x)^2}{2}$

12. $\underline{-\frac{1}{3} \sqrt{(1-x^2)^3} + C}$

析: $\therefore \int x f(x) dx = \arcsin x + C.$

$\therefore x f(x) = (\arcsin x + C)' = \frac{1}{\sqrt{1-x^2}}$ $\therefore f(x) = \frac{1}{x\sqrt{1-x^2}}$

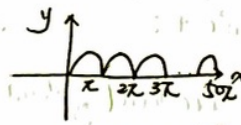
$\therefore \int \frac{1}{x\sqrt{1-x^2}} dx = \int x \sqrt{1-x^2} dx = \frac{1}{2} \int \sqrt{1-x^2} dx^2 = -\frac{1}{2} \int (1-x^2)^{\frac{3}{2}} d(1-x^2) = -\frac{1}{3} (1-x^2)^{\frac{3}{2}} + C.$

13. $\underline{\frac{1}{4} [f(x^2)]^2 + C}$

析: $\int x f(x^2) \cdot f'(x^2) dx = \frac{1}{2} \int f(x^2) f'(x^2) d(x^2) = \frac{1}{2} \int f(x^2) df(x^2).$

(令 $t = f(x^2)$) $\frac{1}{2} \int t dt = \frac{1}{4} t^2 + C = \frac{1}{4} [f(x^2)]^2 + C.$

14. $\underline{200}$

析: $\int_{-50\pi}^{50\pi} |\sin x| dx = 2 \int_0^{50\pi} \sin x dx$ 
 $= 2 \times 50 \int_0^{\pi} (\sin x) dx = 100 \cdot [-\cos x]_0^{\pi} = 200.$

或：运用公式：若 $f(x)$ 是周期为 T 的函数，则 (教材 P₂₅₀)

$$(1) \int_a^{a+T} f(x) dx = \int_0^T f(x) dx.$$

$$(2) \int_a^{a+nT} f(x) dx = n \int_0^T f(x) dx.$$

$$\int_{-50\pi}^{50\pi} |\sin x| dx = \int_{-50\pi}^{50\pi} |\sin x| dx = 100 \int_0^{\pi} |\sin x| dx = 100 \cdot [-\cos x]_0^{\pi} = 200.$$

($|\sin x|$ 是最小正周期为 π 的连续函数).

15. $\frac{\pi}{4}$

析： $\int_0^1 \sqrt{2x-x^2} dx$ 表示圆在 $(1,0)$ 半径为 1 的圆 = $\frac{1}{4}$ 面积.

令 $\sqrt{2x-x^2} = y$, 则 $y \geq 0$. $2x-x^2 = y^2$. 即 $x^2-2x+y^2=0$. 也即 $(x-1)^2+y^2=1$.

$$\therefore \int_0^1 \sqrt{2x-x^2} dx = \frac{1}{4} \pi \cdot 1^2 = \frac{\pi}{4}. \quad \text{即利用定积分的几何意义求解.}$$

16. $\frac{\pi}{8}$

析： $\because x^3 \cos^2 x$ 是奇函数， $\sin^2 x \cos^2 x$ 是偶函数

$$\begin{aligned} \therefore \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (x^3 + \sin^2 x) \cos^2 x dx &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x \cos^2 x dx = 2 \int_0^{\frac{\pi}{2}} \sin^2 x \cos^2 x dx \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{1-\cos 4x}{2} dx = \frac{1}{4} \left[x - \frac{\sin 4x}{4} \right]_0^{\frac{\pi}{2}} = \frac{\pi}{8} \end{aligned}$$

17. $\frac{\pi}{3}$

析： $\because \frac{x^3-3x}{\sqrt{1-x^2}}$ 在 $[-\frac{1}{2}, \frac{1}{2}]$ 上是奇函数.

$$\therefore \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{x^3-3x}{\sqrt{1-x^2}} dx = 0 + 2 \int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^2}} = 2 \cdot [\arcsin x]_0^{\frac{1}{2}} = 2 \cdot \frac{\pi}{6} = \frac{\pi}{3}.$$

18. $\frac{2}{3}$

析： $\int_{-1}^1 (x + \sqrt{1-x^2}) dx = \int_{-1}^1 (x^2 + 1 - x^2 + 2x\sqrt{1-x^2}) dx = \int_{-1}^1 1 dx + 0$ ($\because 2x\sqrt{1-x^2}$ 在 $[-1,1]$ 上是奇函数).

$$= 2.$$

19. $-\frac{1}{2}$

析：令 $x-1=t$, 则 $dx=dt$. $\int_{\frac{1}{2}}^2 f(x-1) dx = \int_{-\frac{1}{2}}^1 f(t) dt = \int_{-\frac{1}{2}}^{\frac{1}{2}} t e^{t^2} dt + \int_{\frac{1}{2}}^1 (1-t) dt$.

$$= 0 + [x]_{\frac{1}{2}}^1 = -\frac{1}{2}. \quad (\because t e^{t^2} \text{ 在 } [-\frac{1}{2}, \frac{1}{2}] \text{ 上是奇函数, 所以其定积分为 } 0)$$

20. 0

析：定积分 $\int_a^b \sin(x+1) dx$ 为常数，所以其导数为 0

21. 1.

析: $\because F(x) = \int_0^x \frac{\sin t}{t} dt \therefore F'(x) = \frac{\sin x}{x} (x \neq 0)$.

$\therefore F'(0)$ 需根据导数定义来求.

$$F'(0) = \lim_{\Delta x \rightarrow 0} \frac{F(0+\Delta x) - F(0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\int_0^{\Delta x} \frac{\sin t}{t} dt - 0}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\frac{\sin \Delta x}{\Delta x}}{1} = 1.$$

22. $\frac{3x^2}{\sqrt{1+x^{12}}} - \frac{2x}{\sqrt{1+x^8}}$

析: $\frac{d}{dx} \int_{x^2}^{x^3} \frac{dt}{\sqrt{1+t^4}} = \frac{(x^3)'}{\sqrt{1+(x^3)^4}} - \frac{(x^2)'}{\sqrt{1+(x^2)^4}} = \frac{3x^2}{\sqrt{1+x^{12}}} - \frac{2x}{\sqrt{1+x^8}}$.

23. 0.

析: $\int_{x^2}^{x^3} \frac{dt}{\sqrt{1+t^4}}$ 是一个关于 x 的函数, 与求导问题中二自变量 (t) 无关.

24. $2t \cos t^2$.

析: $\frac{d}{dt} \int_0^{t^2} \cos x dx = \cos t^2 \cdot (t^2)' = 2t \cos t^2$.

积分变限函数求导公式: $\frac{d}{dx} \int_{\varphi(x)}^{\psi(x)} f(t) dt = f(\psi(x)) \cdot \psi'(x) - f(\varphi(x)) \varphi'(x)$.
(一般形式).

要记住: 积分变限函数为对称, 后求极值, 求极值. 这题求导与二结合 (见 P44 ex1-ex6).

二. 选择题.

1. C.

析: 令 $\sqrt{x} = t$, 则 $x = t^2 (t \geq 0)$. $dx = 2t dt$.

$$\begin{aligned} \int_0^{+\infty} a e^{\sqrt{x}} dx &= 2a \int_0^{+\infty} t e^{-t} dt = 2a \int_0^{+\infty} (-t) d e^{-t} = -2a [t e^{-t}]_0^{+\infty} + 2a \int_0^{+\infty} e^{-t} dt \\ &= -2a \lim_{t \rightarrow +\infty} t e^{-t} - 2a [e^{-t}]_0^{+\infty} \end{aligned}$$

因为 $\lim_{t \rightarrow +\infty} e^{-t} = 0$, $\lim_{t \rightarrow +\infty} t e^{-t} = \lim_{t \rightarrow +\infty} \frac{t}{e^t} = \lim_{t \rightarrow +\infty} \frac{1}{e^t} = 0$
 $\therefore \int_0^{+\infty} a e^{\sqrt{x}} dx = 0 - 2a \cdot 0 + 2a = 2a$. 由题知 $2a = 1$. $\therefore a = \frac{1}{2}$.

2. B.

析: 应用不等式 $\tan x > x (x > 0)$, $\therefore \frac{\tan x}{x} > 1$, $\frac{x}{\tan x} < 1$.

$$\therefore I_1 = \int_0^{\frac{\pi}{4}} \frac{\tan x}{x} dx > \int_0^{\frac{\pi}{4}} dx = \frac{\pi}{4}, \quad I_2 = \int_0^{\frac{\pi}{4}} \frac{x}{\tan x} dx < \int_0^{\frac{\pi}{4}} dx < \frac{\pi}{4}.$$

$\therefore I_1 > I_2$ 且 $I_2 < \frac{\pi}{4}$. 由此排除 A, C, D 项.

3. A.

析: A) $\frac{x}{x^2+1}$ 在 $[0, 5]$ 上连续, 且有原函数 $\frac{1}{2} \ln(x^2+1)$. 故可直接运用 N-L 公式.

B) $\frac{x}{\sqrt{1-x^2}}$ 在积分区间二端无意义, 且在区间上无奇点, 故不能积分.

C) $\frac{1}{x \ln x}$ 在 $[e^1, e]$ 中有无穷间断点, 故不能积分. D) $\int_1^{+\infty} \frac{dx}{x}$ 发散. B, C, D 都不能用 N-L, 不能直接用 N-L.

4. D

析: 对于M: $\frac{\sin x}{1+x^2} \cos^4 x$ 是 $[-\frac{\pi}{2}, \frac{\pi}{2}]$ 上的奇函数, $\therefore M=0$.

对于N: $\sin^3 x$ 是 $[-\frac{\pi}{2}, \frac{\pi}{2}]$ 上的奇函数, $\cos^4 x$ 是偶函数, 且 $\cos^4 x > 0$, $\therefore N = 2 \int_0^{\frac{\pi}{2}} \cos^4 x dx > 0$.

对于P: 因为 $x \sin^3 x$ 是奇函数, $-\cos^4 x$ 是偶函数, 且 $-\cos^4 x < 0$, $\therefore P = 2 \int_0^{\frac{\pi}{2}} (-\cos^4 x) dx < 0$.

$\therefore P < M < N$. 故选D.

5. A

析: $F'(x) = f(\ln x) \cdot (\ln x)' - f(\frac{1}{x}) \cdot (\frac{1}{x})' = \frac{1}{x} f(\ln x) + \frac{1}{x^2} f(\frac{1}{x})$. 故选A.

6. D

析: $I = t \int_0^{\frac{s}{t}} f(tx) dx$. 积分变量是 x , 所以在求定积分过程中 t 与 s 都是常量.

令 $tx = u$, 则 $x = \frac{u}{t}$, $dx = \frac{1}{t} du$. (注意, 这里 t 是常量, 可以理解为一个常数).
当 $x=0$ 时, $u=0$; 当 $x=\frac{s}{t}$ 时, $u=s$.

$\therefore I = t \int_0^s f(u) \cdot \frac{du}{t} = t \cdot \frac{1}{t} \int_0^s f(u) du = \int_0^s f(u) du$. 所以定积分结果只与 s 有关.

7. C

析: $I = \int_0^a \frac{f(x)}{f(x)+f(a-x)} dx \xrightarrow{\text{令 } a-x=t} \int_a^0 \frac{f(a-t)}{f(a-t)+f(t)} dt = \int_0^a \frac{f(a-t)}{f(a-t)+f(t)} dt = \int_0^a \frac{f(a-x)}{f(a-x)+f(x)} dx$

$\therefore 2I = \int_0^a \frac{f(x)}{f(x)+f(a-x)} dx + \int_0^a \frac{f(a-x)}{f(a-x)+f(x)} dx = \int_0^a dx = a$. $\therefore I = \frac{a}{2}$.

8. A

析: 由题知, $\int f(x) dx = e^{-x} + C \Leftrightarrow f(x) = -e^{-x}$.

$\therefore f(\ln x) = -e^{-\ln x} = -e^{\ln \frac{1}{x}} = -\frac{1}{x}$.

$\therefore \int_1^{\sqrt{2}} \frac{1}{x^2} f(\ln x) dx = \int_1^{\sqrt{2}} -\frac{dx}{x^3} = -\frac{1}{2} [x^{-2}]_1^{\sqrt{2}} = -\frac{1}{4}$.

9. B

析: $\because 0 \leq x \leq 1, \therefore 1 \leq \sqrt{1+x} \leq \sqrt{2}, \therefore \frac{\sqrt{2}}{2} \leq \frac{1}{\sqrt{1+x}} \leq 1$

$\frac{\sqrt{2}}{2} \int_0^1 x^4 dx \leq I \leq \int_0^1 x^4 dx, \therefore \int_0^1 x^4 dx = \frac{1}{5}$.

$\therefore \frac{\sqrt{2}}{10} \leq I \leq \frac{1}{5}$.

10. A

析: $\because dw = F \cdot ds = k s ds$.

$\therefore W = \int_0^6 k s ds = \frac{k}{2} [s^2]_0^6 = 18k \text{ (N} \cdot \text{cm)} = 0.18k \text{ (J)}$.

11. A.

析: $\int_1^{+\infty} \frac{dx}{x^2+2} = \frac{1}{2} \int_1^{+\infty} \frac{dx}{(\frac{x}{\sqrt{2}})^2+1} = \frac{\sqrt{2}}{2} \int_1^{\frac{\sqrt{2}}{2}} \frac{d\frac{x}{\sqrt{2}}}{(\frac{x}{\sqrt{2}})^2+1} = \frac{\sqrt{2}}{2} [\arctan \frac{x}{\sqrt{2}}]_1^{+\infty}$

A): $\int_1^{+\infty} \frac{dx}{x^2+2} = \frac{\sqrt{2}}{2} \lim_{x \rightarrow +\infty} (\arctan \frac{x}{\sqrt{2}}) - \frac{\sqrt{2}}{2} \arctan \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} (\frac{\pi}{2} - \arctan \frac{1}{\sqrt{2}})$

B): $\int_e^{+\infty} \frac{dx}{x \sqrt{\ln x+2}} = \int_e^{+\infty} \frac{d \ln x}{\sqrt{\ln x+2}} = \int_e^{+\infty} (\ln x+2)^{-\frac{1}{2}} d(\ln x+2) = 2[(\ln x+2)^{\frac{1}{2}}]_e^{+\infty}$
 $= 2(\lim_{x \rightarrow +\infty} \sqrt{\ln x+2} - \sqrt{3})$ $\because \lim_{x \rightarrow +\infty} \sqrt{\ln x+2} = +\infty$

$\therefore$ 该反常积分发散.

C): $\int_{-\infty}^{+\infty} \sin x dx = \int_{-\infty}^0 \sin x dx + \int_0^{+\infty} \sin x dx$

$\therefore \int_0^{+\infty} \sin x dx = [-\cos x]_0^{+\infty}$ 由于 $\lim_{x \rightarrow +\infty} \cos x$ 不存在.

$\therefore$ 该反常积分发散.

D): $\int_0^1 \frac{dx}{x-1} = [\ln|x-1|]_0^1 = \lim_{x \rightarrow 1^-} \ln|x-1| = -\infty$

$\therefore$ 该反常积分发散.

补漏掉的填空题(11)

二、填空题

11. $x f(-x^2)$

析: 此题略有难度. 注意: 对变量 x 求导, 在初积函数 $f(t^2-x^2)$ 中含 x , 但无法提取. 所以不能直接套用 $\frac{d}{dx} \int_{\varphi(x)}^{\psi(x)} f(t) dt = f(\psi(x))\psi'(x) - f(\varphi(x))\varphi'(x)$.

先处理 $\int_0^x t f(t^2-x^2) dt$.

令 $t^2-x^2=u$, 则 $2t dt = du$. 即 $t dt = \frac{du}{2}$ (注意: 在定积分中 x 为常数).

$\int_0^x t f(t^2-x^2) dt = \int_{0-x^2}^{x^2-x^2} \frac{1}{2} f(u) du = \frac{1}{2} \int_{-x^2}^0 f(t) dt$

注意看右侧的定积分: 初积函数中不再包含求导的变量 x .

$\therefore \frac{d}{dx} \int_0^x t f(t^2-x^2) dt = \frac{d}{dx} [\frac{1}{2} \int_{-x^2}^0 f(t) dt] = -\frac{1}{2} f(-x^2) \cdot (-x^2)' = x f(-x^2)$

三. 解答题.

1. 解: 令 $x = t^6$ ($t > 0$), 则 $t = \sqrt[6]{x}$, $\sqrt{x} = t^3$, $\sqrt[3]{x} = t^2$, $dx = 6t^5 dt$.

$$\begin{aligned}\int \frac{dx}{\sqrt{x}(1+\sqrt[3]{x})} &= \int \frac{6t^5 dt}{t^3(1+t^2)} = 6 \int \frac{t^2 dt}{1+t^2} = 6 \int \frac{(1+t^2)-1}{1+t^2} dt = 6 \int [1 - \frac{1}{1+t^2}] dt \\ &= 6(t - \arctan t) + C = 6(\sqrt[6]{x} - \arctan \sqrt[6]{x}) + C.\end{aligned}$$

2. 解: $\because \frac{1}{x^2(1+x^2)} = \frac{1}{x^2+1} - \frac{x^2-1}{x^4} = \frac{1}{x^2+1} - \frac{1}{x^2} + \frac{1}{x^4}$.

$$\therefore \text{原式} = \int \frac{dx}{x^2+1} - \int \frac{1}{x^2} dx + \int x^{-4} dx = \arctan x + \frac{1}{x} - \frac{1}{3x^3} + C.$$

3. 解: $\because \frac{\sin x}{1+\sin x} = \frac{(1+\sin x)-1}{1+\sin x} = 1 - \frac{1}{1+\sin x} = 1 - \frac{1-\sin x}{(1+\sin x)(1-\sin x)} = 1 - \frac{1-\sin x}{\cos^2 x}$ 分部积分

$$= 1 - \frac{1}{\cos^2 x} + \frac{\sin x}{\cos^2 x} = 1 - \sec^2 x + \sec x \tan x.$$

$$\therefore \text{原式} = \int (1 - \sec^2 x + \sec x \tan x) dx = x - \tan x + \sec x + C.$$

4. 解: 令 $e^x = t$ ($t > 0$), 则 $x = \ln t$, $dx = \frac{dt}{t}$.

$$\begin{aligned}\text{原式} &= \int \frac{\arctan t}{t^2} \cdot \frac{dt}{t} = \int \arctan t \cdot t^{-3} dt = \int \arctan t d(-\frac{t^{-2}}{2}) \\ &= -\frac{t^{-2}}{2} \arctan t - \int (-\frac{t^{-2}}{2}) \cdot \frac{dt}{1+t^2} = -\frac{\arctan t}{2t^2} + \frac{1}{2} \int \frac{dt}{t^2(1+t^2)} \\ &\stackrel{\text{分部积分法}}{=} -\frac{\arctan t}{2t^2} + \frac{1}{2} \int (\frac{1}{t^2} - \frac{1}{1+t^2}) dt = -\frac{\arctan t}{2t^2} - \frac{1}{2t} - \frac{1}{2} \arctan t + C \\ &= -\frac{1}{2} (e^{-2x} \arctan e^x + e^{-x} + \arctan e^x) + C.\end{aligned}$$

5. 解: 令 $\sqrt{\frac{1+x}{x}} = t$ (x>0, t>1) 则 $x = \frac{1}{t^2-1}$ 分部积分法

$$\begin{aligned}\text{原式} &= \int \ln(1+t) d \frac{1}{t^2-1} = \frac{\ln(t+1)}{t^2-1} - \int \frac{dt}{(t^2-1)(1+t)} \\ &\because \frac{1}{(t^2-1)(1+t)} = \frac{1}{(1+t)^2(t-1)} = \frac{1}{4} \left[\frac{1}{t-1} - \frac{t+3}{(t+1)^2} \right] = \frac{1}{4} \left[\frac{1}{t-1} - \frac{t+1+2}{(t+1)^2} \right] \\ &= \frac{1}{4} \left[\frac{1}{t-1} - \frac{1}{t+1} - \frac{2}{(t+1)^2} \right]\end{aligned}$$

$$\begin{aligned}\therefore \text{原式} &= \frac{\ln(t+1)}{t^2-1} - \frac{1}{4} \int \left[\frac{1}{t-1} - \frac{1}{t+1} - \frac{2}{(t+1)^2} \right] dt \\ &= \frac{\ln(t+1)}{t^2-1} - \frac{1}{4} \ln(t-1) + \frac{1}{4} \ln(t+1) + \frac{1}{2} \int \frac{d(t+1)}{(t+1)^2} \\ &= \frac{\ln(t+1)}{t^2-1} + \frac{1}{4} \ln \frac{t+1}{t-1} - \frac{1}{2(t+1)} + C \\ &= x \ln(1+\sqrt{\frac{1+x}{x}}) + \frac{1}{2} \ln(\sqrt{1+x} + \sqrt{x}) + \frac{x}{2} - \frac{\sqrt{x(1+x)}}{2} + C.\end{aligned}$$

6. 解: $\because \cos A \sin B = -\frac{1}{2} [\cos(A+B) - \cos(A-B)]$

$$\therefore \sin x \cdot \sin 2x = -\frac{1}{2} (\cos 3x - \cos x)$$

$$\therefore \int \sin x \sin 2x \cdot \sin 3x dx = \frac{1}{2} \int \cos x \cdot \sin 3x dx - \frac{1}{2} \int \cos 3x \sin 3x dx$$

$$\because \cos A \sin B = \frac{1}{2} (\sin(A+B) - \sin(A-B))$$

$$\therefore \cos x \cdot \sin 3x = \frac{1}{2} (\sin 4x + \sin 2x)$$

$$\therefore \int \sin x \sin 2x \sin 3x dx = \frac{1}{4} \int (\sin 4x + \sin 2x) dx - \frac{1}{4} \int \sin 3x \cos 3x dx$$

$$= -\frac{1}{16} \cos 4x - \frac{1}{8} \cos 2x - \frac{1}{12} \sin^2 3x + C$$

解法: $\because \sin x \sin 3x = -\frac{1}{2} (\cos 4x - \cos 2x)$

$$\therefore \int \sin x \sin 2x \sin 3x dx = -\frac{1}{2} \int (\cos 4x - \cos 2x) \sin 2x dx$$

$$= \frac{1}{4} \int (\cos 4x - \cos 2x) d \cos 2x = \frac{1}{4} \int (2 \cos^2 2x - \cos 2x - 1) d \cos 2x$$

$$= \frac{1}{6} \cos^3 2x - \frac{1}{8} \cos^2 2x - \frac{1}{4} \cos 2x + C$$

7. 解: 令 $\sqrt{\frac{1-x}{1+x}} = t, (t > 0)$. 则 $x = \frac{1-t^2}{t^2+1}, dx = \frac{-4t}{(t^2+1)^2} dt$

$$\text{原式} = \int \frac{t^2+1}{1-t^2} \cdot t \cdot \frac{-4t}{(t^2+1)^2} dt$$

$$= \int \frac{4t^2}{(t^2-1)(t^2+1)} dt = \int \left(\frac{2}{t^2-1} + \frac{2}{t^2+1} \right) dt$$

$$= \int \left(\frac{1}{t-1} - \frac{1}{t+1} + \frac{2}{t^2+1} \right) dt$$

$$= \ln|t-1| - \ln|t+1| + 2 \arctan t + C$$

$$= \ln \left| \frac{t-1}{t+1} \right| + 2 \arctan t + C$$

$$= \ln \left| \frac{\sqrt{1-x} - \sqrt{1+x}}{\sqrt{1-x} + \sqrt{1+x}} \right| + 2 \arctan \sqrt{\frac{1-x}{1+x}} + C$$

8. 解: 令 $\sqrt[6]{1+x} = t, (t > 0)$. 则 $x = t^6 - 1, dx = 6t^5 dt$

$$\text{原式} = \int \frac{6t^5 dt}{t^3 + t^2} = 6 \int \frac{t^3 dt}{t+1} = 6 \int \left(t^2 - t + \frac{1}{t+1} \right) dt = 2t^3 - 3t^2 + 6t - 6 \ln|t+1| + C$$

$$= 2\sqrt[6]{1+x} - 3\sqrt[3]{1+x} + 6\sqrt[6]{x+1} - 6 \ln(\sqrt[6]{1+x} + 1) + C$$

9. 解: $\int \frac{\arctan x}{x^2(1+x^2)} dx = \int \frac{\arctan x}{x^2} dx - \int \frac{\arctan x}{1+x^2} dx = \int \arctan x d(-\frac{1}{x}) - \int \arctan x d \arctan x$

$$= -\frac{1}{x} \arctan x + \int \frac{dx}{x(1+x^2)} - \frac{1}{2} (\arctan x)^2$$

$$= -\frac{\arctan x}{x} - \frac{1}{2} (\arctan x)^2 + \int \frac{dx}{x} - \int \frac{x dx}{1+x^2}$$

$$= -\frac{\arctan x}{x} - \frac{1}{2} (\arctan x)^2 + \ln|x| - \frac{1}{2} \int \frac{dx^2}{x^2+1}$$

$$= -\frac{\arctan x}{x} - \frac{1}{2} (\arctan x)^2 + \ln|x| - \frac{1}{2} \ln(x^2+1) + C$$

(或写成: $-\frac{\arctan x}{x} - \frac{1}{2} (\arctan x)^2 + \frac{1}{2} \ln \frac{x^2}{x^2+1} + C$.)

10. 解: ① 当 $x < 0$ 时, $\int f(x) dx = \int dx = x + C_1$
 ② 当 $0 \leq x \leq 1$ 时, $\int f(x) dx = \int (x+1) dx = \frac{x^2}{2} + x + C_2$
 ③ 当 $x > 1$ 时, $\int f(x) dx = \int 2x dx = x^2 + C_3$
 由于原函数在 $x=0, x=1$ 处连续, 可知

$$C_1 = C_2; \quad \frac{1}{2} + 1 + C_2 = 1 + C_3 \quad \text{即} \quad C_3 = C_2 + \frac{1}{2}$$

令 $C_1 = C_2 = C_3 - \frac{1}{2} = C$, 则

$$\int f(x) dx = \begin{cases} x + C, & \text{当 } x < 0 \text{ 时,} \\ \frac{x^2}{2} + x + C, & \text{当 } 0 \leq x \leq 1 \text{ 时,} \\ x^2 + \frac{1}{2} + C, & \text{当 } x > 1 \text{ 时.} \end{cases}$$

11. 解: 令 $\arcsin e^x = t$. ($-\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$). 则 $e^x = \sin t, x = \ln \sin t$

$$dx = \frac{\cos t}{\sin t} dt$$

$$\therefore \text{原式} = \int \frac{t}{\sin t} \cdot \frac{\cos t}{\sin t} dt = \int t \cdot \cot t \cdot \csc t dt = \int t d(-\csc t)$$

$$= -t \csc t + \int \csc t dt = -t \csc t + \ln |\csc t - \cot t| + C$$

$$= -\frac{\arcsin e^x}{e^x} + \ln \left| \frac{1}{e^x} - \frac{\sqrt{1-e^{2x}}}{e^x} \right| + C$$

$$= -\frac{\arcsin e^x}{e^x} + \ln(1 - \sqrt{1-e^{2x}}) - x + C$$

12. 解: 原式 $= \int \ln \sin x d(-\cot x) = -\cot x \cdot \ln \sin x + \int \cot x \cdot \frac{\cos x}{\sin x} dx$

$$= -\cot x \ln \sin x + \int \cot^2 x dx = -\cot x \ln \sin x + \int (\csc^2 x - 1) dx$$

$$= -\cot x \ln \sin x - \cot x - x + C$$

$$\begin{aligned}
 13. \text{ 解: 原式} &= \int \frac{dx}{2\sin x \cos x + 2\sin x} = \int \frac{dx}{2\sin x (\cos x + 1)} = \int \frac{dt}{2 \cdot 2\sin \frac{x}{2} \cos \frac{x}{2} \cdot 2\cos^2 \frac{x}{2}} \\
 &= \frac{1}{4} \int \frac{d\frac{x}{2}}{\sin \frac{x}{2} \cos^3 \frac{x}{2}} = \frac{1}{4} \int \frac{\frac{d\frac{x}{2}}{\cos \frac{x}{2}}}{\tan \frac{x}{2}} = \frac{1}{4} \int \frac{\sec^2 \frac{x}{2} \cdot \sec^2 \frac{x}{2} d\frac{x}{2}}{\tan \frac{x}{2}} \\
 &= \frac{1}{4} \int \frac{\sec^2 \frac{x}{2} d\tan \frac{x}{2}}{\tan \frac{x}{2}} = \frac{1}{4} \int \frac{\tan^2 \frac{x}{2} + 1}{\tan \frac{x}{2}} d\tan \frac{x}{2} = \frac{1}{4} \int \left(\tan \frac{x}{2} + \frac{1}{\tan \frac{x}{2}} \right) d\tan \frac{x}{2} \\
 &= \frac{1}{8} (\tan \frac{x}{2})^2 + \frac{1}{4} \ln |\tan \frac{x}{2}| + C.
 \end{aligned}$$

说明: 出题时没有考虑到各类方法, 所以像三角代换等较复杂的题型请自行围绕课后习题练习。

$$\begin{aligned}
 14. \text{ 解: } \int_0^\pi f(x) dx &= [x f(x)]_0^\pi - \int_0^\pi x df(x) \\
 \because f(0) &= \int_\pi^x \frac{\sin t}{t} dt \quad \therefore f(\pi) = 0, \quad f'(x) = \frac{\sin x}{x} \\
 \therefore \int_0^\pi f(x) dx &= - \int_0^\pi x \cdot \frac{\sin x}{x} dx = [\cos x]_0^\pi = -2.
 \end{aligned}$$

$$\begin{aligned}
 15. \text{ 解: } \int_0^\pi \sqrt{\cos^2 x - \cos^4 x} dx &= \int_0^\pi \sqrt{\cos^2 x \cdot \sin^2 x} dx = \int_0^\pi |\cos x| \sin x dx = \int_0^{\frac{\pi}{2}} \cos x \sin x dx - \int_{\frac{\pi}{2}}^\pi \cos x \sin x dx \\
 &= \frac{1}{2} [\sin^2 x]_0^{\frac{\pi}{2}} - \frac{1}{2} [\sin^2 x]_{\frac{\pi}{2}}^\pi = 1.
 \end{aligned}$$

16. 解: 由题可知 $0 \leq t \leq 1$.

$$\therefore \text{① 当 } x < 0 \text{ 时, } x < t. \quad \int_0^1 t|t-x| dt = \int_0^1 t(t-x) dt = \int_0^1 (t^2 - tx) dt = \left[\frac{t^3}{3} - \frac{x}{2} t^2 \right]_0^1 = \frac{1}{3} - \frac{x}{2}.$$

$$\text{② 当 } x > 1 \text{ 时, } x > t. \quad \int_0^1 t|t-x| dt = \int_0^1 t(x-t) dt = \left[\frac{xt^2}{2} - \frac{t^3}{3} \right]_0^1 = \frac{x}{2} - \frac{1}{3}.$$

$$\begin{aligned}
 \text{③ 当 } 0 \leq x \leq 1 \text{ 时, } \int_0^1 t|t-x| dt &= \int_0^x t|t-x| dt + \int_x^1 t|t-x| dt \\
 &= \int_0^x t(x-t) dt + \int_x^1 t(t-x) dt \\
 &= \left[\frac{xt^2}{2} - \frac{t^3}{3} \right]_0^x + \left[\frac{t^3}{3} - \frac{x}{2} t^2 \right]_x^1 \\
 &= \frac{x^3}{3} - \frac{x^2}{2} + \frac{1}{3}.
 \end{aligned}$$

$$\therefore \int_0^1 t|t-x| dt = \begin{cases} \frac{1}{3} - \frac{x}{2}, & x < 0, \\ \frac{x^3}{3} - \frac{x^2}{2} + \frac{1}{3}, & 0 \leq x \leq 1 \\ \frac{x}{2} - \frac{1}{3}, & x > 1. \end{cases}$$

✶

17. 解: $I = \int_{-\pi}^0 \frac{x \sin x \cdot \arctan e^x}{1 + \cos^2 x} dx + \int_0^{\pi} \frac{x \sin x \cdot \arctan e^x}{1 + \cos^2 x} dx$

$$= \int_{-\pi}^0 \frac{(-u) \sin u \cdot \arctan e^{-u}}{1 + \cos^2 u} d(-u) + \int_0^{\pi} \frac{x \sin x \cdot \arctan e^x}{1 + \cos^2 x} dx$$

$$= \int_0^{\pi} \frac{u \sin u \cdot \arctan e^{-u}}{1 + \cos^2 u} du + \int_0^{\pi} \frac{x \sin x \cdot \arctan e^x}{1 + \cos^2 x} dx$$

$$= \int_0^{\pi} \frac{x \sin x \cdot \arctan e^{-x}}{1 + \cos^2 x} dx + \int_0^{\pi} \frac{x \sin x \cdot \arctan e^x}{1 + \cos^2 x} dx$$

$$= \int_0^{\pi} (\arctan e^{-x} + \arctan e^x) \cdot \frac{x \sin x}{1 + \cos^2 x} dx.$$

因为 $(\arctan e^{-x} + \arctan e^x)' = \frac{-e^{-x}}{1+e^{-2x}} + \frac{e^x}{1+e^{2x}} = \frac{-e^x}{e^{2x}+1} + \frac{e^x}{1+e^{2x}} = 0$

$$\therefore \arctan e^{-x} + \arctan e^x = C.$$

令 $x=0$, 得 $C = \frac{\pi}{2}$. $\therefore \arctan e^{-x} + \arctan e^x = \frac{\pi}{2}$.

$$\therefore I = \frac{\pi}{2} \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$$

$$= \left(\frac{\pi}{2}\right)^2 \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx \quad \left(\text{因为 } \int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx \right)$$

$$= -\left(\frac{\pi}{2}\right)^2 \int_0^{\pi} \frac{d \cos x}{1 + \cos^2 x} = -\left(\frac{\pi}{2}\right)^2 [\arctan \cos x]_0^{\pi} = \frac{\pi^3}{8}.$$

(说明: 本题在计算定积分过程中属于难题, 前边处理技巧前没有遇到过, 用3个公式)

18. 解: $I = \int_0^{\frac{\pi}{2}} \frac{dx}{1 + \frac{(\sin x)^{\sqrt{3}}}{(\cos x)^{\sqrt{3}}}} = \int_0^{\frac{\pi}{2}} \frac{(\cos x)^{\sqrt{3}}}{(\sin x)^{\sqrt{3}} + (\cos x)^{\sqrt{3}}} dx, \quad \text{令 } x = \frac{\pi}{2} - t \quad - \int_{\frac{\pi}{2}}^0 \frac{(\sin t)^{\sqrt{3}}}{(\cos t)^{\sqrt{3}} + (\sin t)^{\sqrt{3}}} dt$

$$= \int_0^{\frac{\pi}{2}} \frac{(\sin t)^{\sqrt{3}}}{(\sin t)^{\sqrt{3}} + (\cos t)^{\sqrt{3}}} dt = \int_0^{\frac{\pi}{2}} \frac{(\sin x)^{\sqrt{3}}}{(\sin x)^{\sqrt{3}} + (\cos x)^{\sqrt{3}}} dx$$

$$\therefore 2I = \int_0^{\frac{\pi}{2}} \frac{(\cos x)^{\sqrt{3}}}{(\sin x)^{\sqrt{3}} + (\cos x)^{\sqrt{3}}} dx + \int_0^{\frac{\pi}{2}} \frac{(\sin x)^{\sqrt{3}}}{(\sin x)^{\sqrt{3}} + (\cos x)^{\sqrt{3}}} dx = \int_0^{\frac{\pi}{2}} dx = \frac{\pi}{2}.$$

$$\therefore I = \frac{\pi}{4}.$$

(说明: 在定积分求解问题中, 并不总用非子函数代入两端点值二方法, 而总用一些技巧进行转化).

19. 出题形式与题5类似.

解: $\int_0^{\frac{\pi}{2}} \sqrt{\sin x} |\cos x| dx = \int_0^{\frac{\pi}{2}} \sqrt{\sin x} \cos x dx + \int_0^{\frac{\pi}{2}} \sqrt{\sin x} (-\cos x) dx.$

$$= \frac{2}{3} [\sin x]^{\frac{3}{2}} \Big|_0^{\frac{\pi}{2}} - \frac{2}{3} [\sin x]^{\frac{3}{2}} \Big|_0^{\frac{\pi}{2}} = \frac{4}{3}.$$

20. 解: 令 $t = \frac{1}{x}$, 则 $dx = -\frac{1}{t^2} dt$.

$$\text{原式} = \int_1^{\frac{1}{3}} \frac{-\frac{dt}{t^2}}{\frac{1}{t} \cdot \frac{\sqrt{t^2+5t+1}}{t}} = \int_{\frac{1}{3}}^1 \frac{dt}{\sqrt{t^2+5t+1}} = \int_{\frac{1}{3}}^1 \frac{dt}{\sqrt{(t+\frac{5}{2})^2 - \frac{21}{4}}}$$

(令 $x = a \sec t$ (当 $a < x$)

因为 $\int \frac{dx}{\sqrt{x^2-a^2}} = \ln(x + \sqrt{x^2-a^2}) + C$. (这个记不住也没关系, 用三角代换方法再求一遍)

所以 原式 $= [\ln(t + \frac{5}{2} + \sqrt{t^2+5t+1})]_{\frac{1}{3}}^1 = \ln \frac{7+2\sqrt{7}}{9}$

21. 解: $\int_0^1 \frac{\ln(1+x)}{(2-x)^2} dx = \int_0^1 \ln(1+x) d(\frac{1}{2-x}) = [\frac{\ln(1+x)}{2-x}]_0^1 - \int_0^1 \frac{dx}{(1+x)(2-x)}$
 $= \ln 2 + \frac{1}{3} \int_0^1 (\frac{1}{x-2} - \frac{1}{x+1}) dx = \ln 2 + \frac{1}{3} [\ln|\frac{x-2}{x+1}|]_0^1 = \frac{\ln 2}{3}$

22. 解: $\therefore f'(2x) dx = \frac{1}{2} df'(2x)$.

$\therefore \int_0^1 x f'(2x) dx = \frac{1}{2} \int_0^1 x df'(2x) = \frac{1}{2} [x \cdot f'(2x)]_0^1 - \frac{1}{2} \int_0^1 f'(2x) dx$
 $= \frac{1}{2} f'(2) - \frac{1}{2} \int_0^1 \frac{1}{2} df(2x) = \frac{1}{2} f'(2) - \frac{1}{4} [f(2x)]_0^1$
 $= \frac{5}{2} - \frac{1}{4} (f(2) - f(0)) = 2$

23. 解: 令 $\sqrt{3x+1} = t$ ($t \geq 0$), 则 $x = \frac{t^2-1}{3}$, $dx = \frac{2t}{3} dt$.

$\int_0^5 \frac{dx}{2x + \sqrt{3x+1}} = \int_1^4 \frac{\frac{2t}{3} dt}{\frac{2(t^2-1)}{3} + t} = \int_1^4 \frac{2t dt}{2t^2 + 3t - 2} = \int_1^4 \frac{2t dt}{(t-2)(2t+1)}$ (从草稿纸上抄过来了!!)
 $= \int_1^4 \frac{1}{5} (\frac{4}{t-2} + \frac{2}{2t+1}) dt = \frac{1}{5} \int_1^4 \frac{4 dt}{t-2} + \frac{1}{5} \int_1^4 \frac{d(2t+1)}{2t+1}$
 $= \frac{4}{5} [\ln|t-2|]_1^4 + \frac{1}{5} [\ln(2t+1)]_1^4 = \frac{4}{5} \ln 2$

23. 解: 令 $\sqrt{3x+1} = t$ ($t \geq 0$), 则 $x = \frac{t^2-1}{3}$, $dx = \frac{2t}{3} dt$

$\int_0^5 \frac{dx}{2x + \sqrt{3x+1}} = \int_1^4 \frac{\frac{2t}{3} dt}{\frac{2(t^2-1)}{3} + t} = \int_1^4 \frac{2t dt}{2t^2 + 3t - 2} = \int_1^4 \frac{2t dt}{(t+2)(2t-1)}$
 $= \frac{2}{5} \int_1^4 (\frac{2}{t+2} + \frac{1}{2t-1}) dt = \frac{2}{5} [2\ln|t+2| + \frac{1}{2} \ln|2t-1|]_1^4 = \frac{1}{5} (4\ln 2 + \ln 7)$

24. 解: $\because \lim_{x \rightarrow 1^-} \frac{x^2 \arcsin x}{\sqrt{1-x^2}} = +\infty \therefore \int_0^1 \frac{x^2 \arcsin x}{\sqrt{1-x^2}} dx$ 为反常积分.

令 $\arcsin x = t$, 则 $x = \sin t$, ($0 \leq t < \frac{\pi}{2}$) $dx = \cos t dt$

$\int_0^1 \frac{x^2 \arcsin x}{\sqrt{1-x^2}} dx = \int_0^{\frac{\pi}{2}} \frac{\sin^2 t \cdot t \cdot \cos t dt}{|\cos t|} = \int_0^{\frac{\pi}{2}} \frac{1-\cos 2t}{2} \cdot t dt = \int_0^{\frac{\pi}{2}} \frac{t}{2} dt - \frac{1}{2} \int_0^{\frac{\pi}{2}} t \cos 2t dt$
 $= [\frac{t^2}{4}]_0^{\frac{\pi}{2}} - \frac{1}{4} \int_0^{\frac{\pi}{2}} t \sin 2t dt = \frac{\pi^2}{16} - \frac{1}{4} [t \sin 2t]_0^{\frac{\pi}{2}} + \frac{1}{4} \int_0^{\frac{\pi}{2}} \sin 2t dt = \frac{\pi^2}{16} - \frac{1}{8} [\cos 2t]_0^{\frac{\pi}{2}}$
 $= \frac{\pi^2}{16} + \frac{1}{4}$

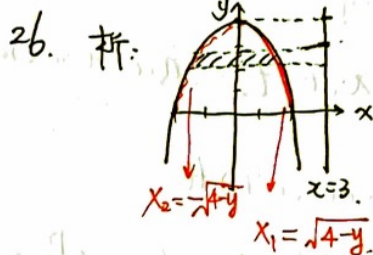
25. 解. 由题可知, $\sqrt{3} \leq x \leq \sqrt{3}$, 且 $y' = \sqrt{3}x^2$

$$\therefore S = \int_{-\sqrt{3}}^{\sqrt{3}} \sqrt{1+y'^2} dx = \int_{-\sqrt{3}}^{\sqrt{3}} \sqrt{1+3x^2} dx = 2 \int_0^{\sqrt{3}} \sqrt{4x^2} dx.$$

令 $x = \frac{2}{\sqrt{3}} \sin t$, 则当 $x=0$ 时, $t=0$, 当 $x=\sqrt{3}$ 时, $t=\frac{\pi}{3}$. $dx = \frac{2}{\sqrt{3}} \cos t dt$.

$$S = 2 \int_0^{\frac{\pi}{3}} 2 \cos t \cdot \frac{2}{\sqrt{3}} \cos t dt = \frac{4}{\sqrt{3}} \int_0^{\frac{\pi}{3}} (\cos^2 t + 1) dt = \left[\frac{2}{\sqrt{3}} \sin 2t + 4t \right]_0^{\frac{\pi}{3}} \\ = \sqrt{3} + \frac{4}{3}\pi.$$

(说明: 关于积分上限函数定义域不清晰, 请阅读教材 P338).



任取 $[y, y+dy]$ 在 $[0, 4]$ 上 (y 轴上).

$$dV_1 = \pi \cdot (3 - \sqrt{4-y})^2 \cdot dy$$

$$dV_2 = \pi [3 - (\sqrt{4-y})]^2 dy.$$

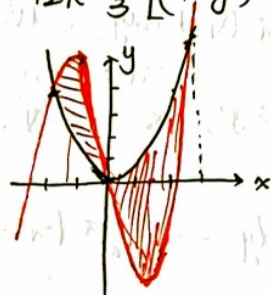
$$dV = dV_2 - dV_1.$$

解: $V = \int_0^4 [\pi(3 + \sqrt{4-y})^2 - \pi(3 - \sqrt{4-y})^2] dy.$

$$= 12\pi \int_0^4 \sqrt{4-y} dy = -12\pi \int_0^4 (4-y)^{\frac{1}{2}} d(4-y)$$

$$= -12\pi \cdot \frac{2}{3} [(4-y)^{\frac{3}{2}}]_0^4 = 64\pi.$$

27. 析: 画草图:



$$S = \int_{-2}^0 [f(x) - g(x)] dx + \int_0^3 [g(x) - f(x)] dx.$$

解: 由 $\begin{cases} y = x^2 - 6x \\ y = x^2 \end{cases}$ 得交点 $(0, 0)$, $(2, 4)$ 和 $(3, 9)$.

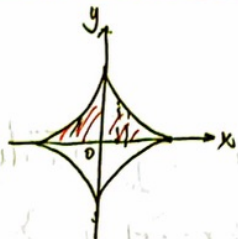
在区间 $[-2, 0]$ 上, 由 $f(1) = 5 > g(1) = 1$ 可知, $f(x) \geq g(x)$, 即 $x^2 - 6x \geq x^2$.
在区间 $[0, 3]$ 上, 由 $f(1) = -5 < g(1) = 1$ 可知, $f(x) \leq g(x)$, 即 $x^2 \geq x^2 - 6x$.

$$\therefore \text{所围面积 } S = \int_{-2}^0 (x^2 - 6x - x^2) dx + \int_0^3 (x^2 - x^2 + 6x) dx.$$

$$= \left[\frac{x^3}{3} - 3x^2 - \frac{x^3}{3} \right]_{-2}^0 + \left[\frac{x^3}{3} - \frac{x^4}{4} + 3x^2 \right]_0^3$$

$$= \frac{253}{12}.$$

28. 析: 星形线



建议: 熟悉教材P31-3/3-2中常见曲线.

解: (1) 由对称性可知.
$$l = 4 \int_0^{\frac{\pi}{2}} \sqrt{[x'(t)]^2 + [y'(t)]^2} dt$$
$$= 4 \int_0^{\frac{\pi}{2}} 3a \cos t \sin t dt = 6a \cdot [\sin^2 t]_0^{\frac{\pi}{2}} = 6a.$$

(2) 实际上由星形线在x轴上方部分绕x轴旋转一周即可.

$$V = \int_{-a}^a \pi y^2 dx = 2 \int_0^a \pi y^2 dx = 2 \int_{-\frac{\pi}{2}}^0 \pi (a \sin^3 t)^2 \cdot 3a \cos^2 t (-\sin t) dt.$$

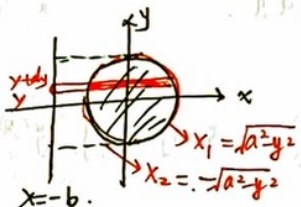
$$= 6\pi a^3 \int_0^{\frac{\pi}{2}} \sin^7 t \cos^2 t dt = 6\pi a^3 \int_0^{\frac{\pi}{2}} \sin^7 t (1 - \sin^2 t) dt$$

$\therefore I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx = \frac{(n-1)!!}{n!!}$ 当n为大于1的正奇数时. (教材P253).

$$\therefore \int_0^{\frac{\pi}{2}} (\sin^7 t - \sin^9 t) dt = \frac{6 \times 4 \times 2}{7 \times 5 \times 3 \times 1} - \frac{8 \times 6 \times 4 \times 2}{9 \times 7 \times 5 \times 3} = \frac{16}{9 \times 7 \times 5}$$

$$\therefore V = 6\pi a^3 \cdot \frac{16}{9 \times 7 \times 5} = \frac{32}{105} \pi a^3.$$

29. 析:



$$dV_1 = \pi (x_1 + b)^2 dy, \quad dV_2 = \pi (x_2 + b)^2 dy.$$

$$x_1 = \sqrt{a^2 - y^2}, \quad x_2 = -\sqrt{a^2 - y^2}.$$

解:
$$V = \int_{-a}^a \pi (\sqrt{a^2 - y^2} + b)^2 dy - \int_{-a}^a \pi (-\sqrt{a^2 - y^2} + b)^2 dy.$$

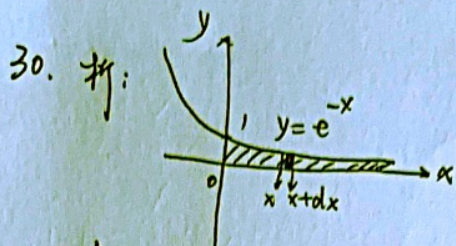
$$= 2\pi \int_0^a [(\sqrt{a^2 - y^2} + b)^2 - (-\sqrt{a^2 - y^2} + b)^2] dy.$$

$$= 8\pi b \int_0^a \sqrt{a^2 - y^2} dy.$$

因为 $\int_0^a \sqrt{a^2 - y^2} dy$ 表示以(0,0)为圆心, 以a为半径之圆在第一象限部分之

面积, 所以 $\int_0^a \sqrt{a^2 - y^2} dy = \frac{\pi a^2}{4}.$

$$\therefore V = 8\pi b \cdot \frac{\pi a^2}{4} = 2\pi^2 a^2 b.$$



解: 当 $x \rightarrow +\infty$ 时, $y \rightarrow 0$. 析也.

$$V = \int_0^{+\infty} \pi \cdot y^2 dx = \int_0^{+\infty} \pi \cdot e^{-2x} dx = -\frac{\pi}{2} [e^{-2x}]_0^{+\infty} \\ = -\frac{\pi}{2} (\lim_{x \rightarrow +\infty} e^{-2x} - 1) = \frac{\pi}{2}.$$

31. 析: 极坐标系下弧长公式. $s = \int_{\alpha}^{\beta} \sqrt{r^2(\theta) + r'^2(\theta)} d\theta$.

解: $\because r(\theta) = a(1 + \cos\theta), \therefore r'(\theta) = -a\sin\theta$.

$$\sqrt{r^2(\theta) + r'^2(\theta)} = \sqrt{a^2(1 + \cos\theta)^2 + a^2\sin^2\theta} = \sqrt{2} \cdot a \sqrt{1 + \cos\theta} = 2a \left| \cos \frac{\theta}{2} \right|.$$

$$\therefore s = \int_0^{2\pi} 2a \left| \cos \frac{\theta}{2} \right| d\theta = 4a \int_0^{2\pi} \left| \cos \frac{\theta}{2} \right| d\frac{\theta}{2} = 4a \int_0^{\pi} |\cos t| dt \\ = 4a \cdot \int_0^{\frac{\pi}{2}} \cos t dt - 4a \int_{\frac{\pi}{2}}^{\pi} \cos t dt = 4a [\sin t]_0^{\frac{\pi}{2}} - 4a [\sin t]_{\frac{\pi}{2}}^{\pi} = 8a.$$

说明: 定积分应用中, 求面积, 弧长, 体积必能会. 尤其求面积和弧长还有各种不同二重积分情形. 具体公式见教材或深讲.